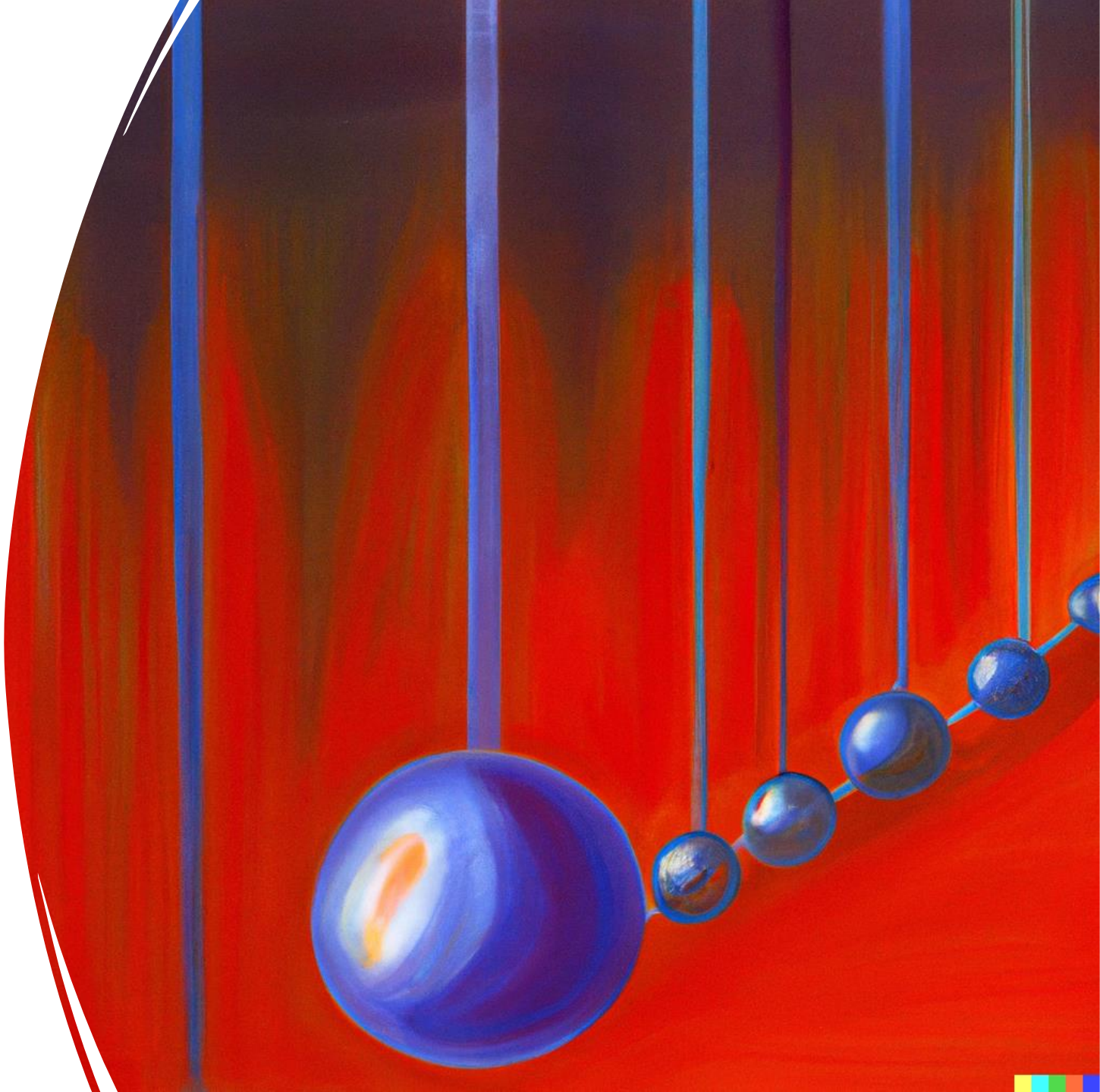
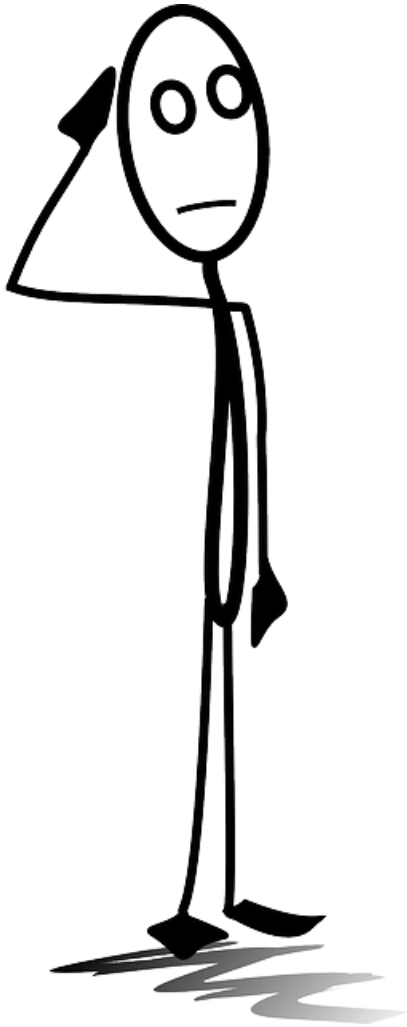
PHYSICS I
WEEK 4

Siddharth Dhomkar
Department of Physics
IIT Madras
sdhomkar@smail.iitm.ac.in
Office: HSB 116B



Week 4: outline

- Simple harmonic motion: Taylor 5.1, 5.2
- Damped Oscillator: Taylor 5.4
- Driven, damped harmonic motion: Taylor 5.5, 5.6

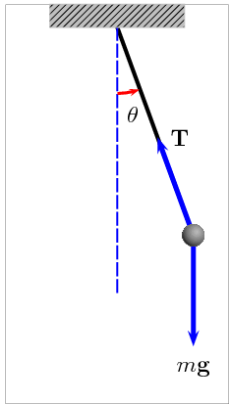


SIMPLE HARMONIC MOTION: SOLUTIONS

$$F_x(x) = -kx$$

$$m\ddot{x} = F_x = -kx$$

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$



$$\omega = \sqrt{\frac{k}{m}}$$

Equation of motion

$$\ddot{\phi} = -\omega^2\phi$$

A second-order, linear, homogeneous differential equation

General Solution

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

C_1 and C_2 depend on the initial conditions (need to be chosen carefully to have real valued $x(t)$)

$$e^{\pm i\omega t} = \cos(\omega t) \pm i\sin(\omega t)$$

Euler's formula

$$x(t) = B_1\cos(\omega t) + B_2\sin(\omega t) \qquad B_1 = C_1 + C_2 \quad \text{and} \quad B_2 = i(C_1 - C_2)$$

$$x(t) = x_0\cos(\omega t)$$ **Initial conditions $x = x_0$ & $v = 0$**

$$x(t) = \frac{v_0}{\omega}\sin(\omega t)$$ **Initial conditions $x = 0$ & $v = v_0$**

Period

$$\tau = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$

SIMPLE HARMONIC MOTION: SOLUTIONS

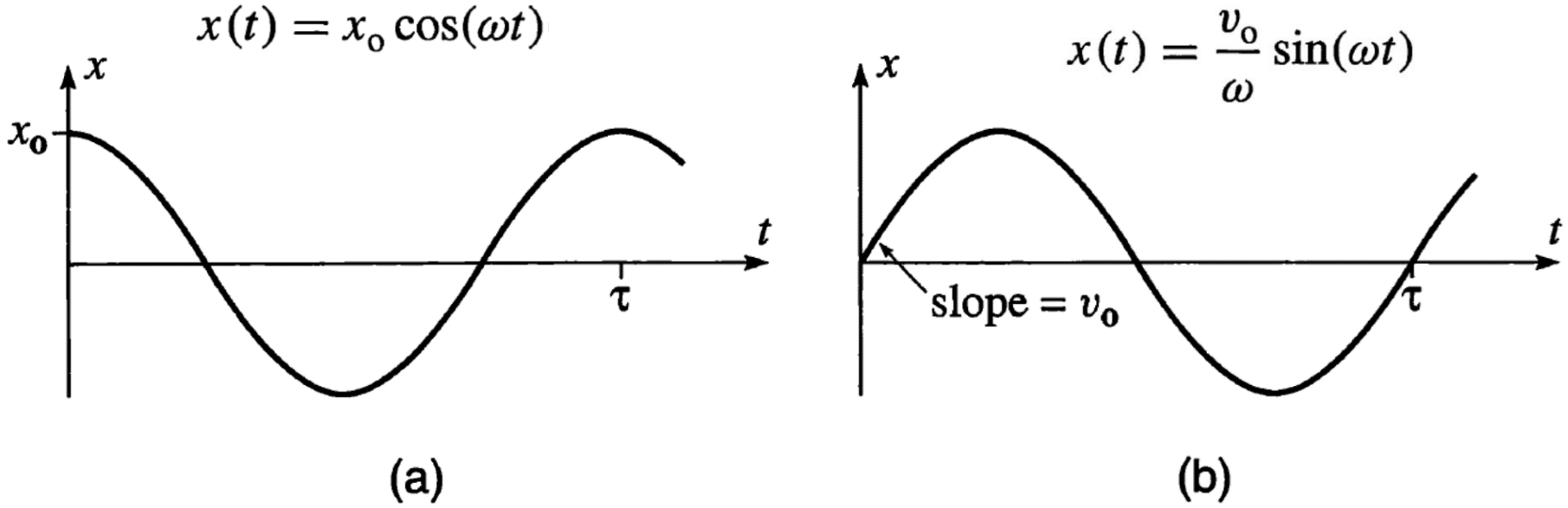


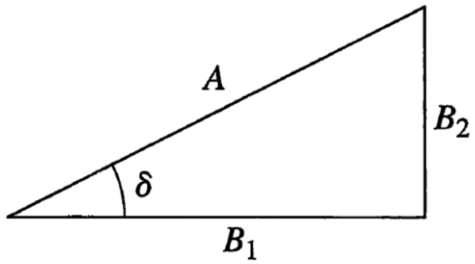
Figure 5.3 **(a)** Oscillations in which the cart is released from x_0 at $t = 0$ follow a cosine curve. **(b)** If the cart is kicked from the origin at $t = 0$, the oscillations follow a sine curve with initial slope v_0. In either case the period of the oscillations is $\tau = 2\pi/\omega = 2\pi\sqrt{m/k}$ and is the same whatever the values of x_0 or v_0.

SIMPLE HARMONIC MOTION: SOLUTIONS

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

$$x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$$



SIMPLE HARMONIC MOTION: SOLUTIONS

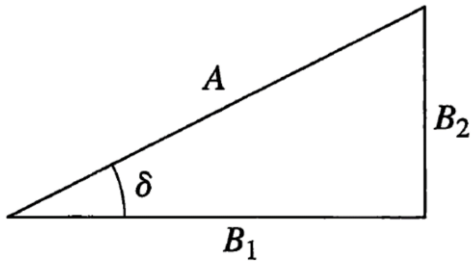
$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

$$x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$$

A phase-shifted cosine

$$x(t) = A\cos(\omega t - \delta)$$



SIMPLE HARMONIC MOTION: SOLUTIONS

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

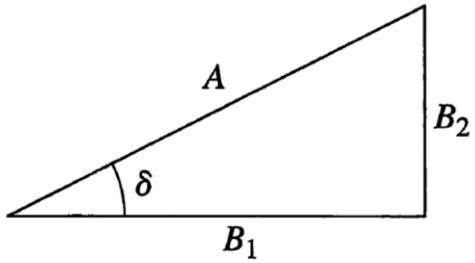
$$x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$$

Real part of the complex exponential

$$x(t) = \mathrm{Re}\, C e^{i\omega t}$$

A phase-shifted cosine

$$x(t) = A \cos(\omega t - \delta)$$



SIMPLE HARMONIC MOTION: SOLUTIONS

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

$$x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$$

Real part of the complex exponential

$$x(t) = \text{Re}\, C e^{i\omega t}$$

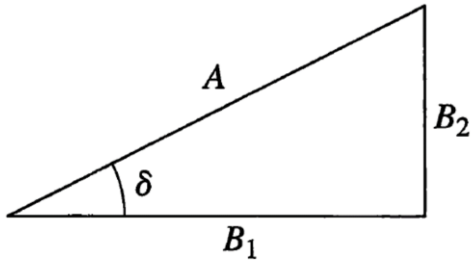
$$= \text{Re}\, A e^{i(\omega t - \delta)}$$

$$C = B_1 - i B_2$$

$$= A e^{-i\delta}$$

A phase-shifted cosine

$$x(t) = A \cos(\omega t - \delta)$$



SIMPLE HARMONIC MOTION: SOLUTIONS

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

$$x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$$

Real part of the complex exponential

$$x(t) = \text{Re}\, C e^{i\omega t}$$

$$= \text{Re}\, A e^{i(\omega t - \delta)}$$

$$C = B_1 - iB_2$$

$$= Ae^{-i\delta}$$

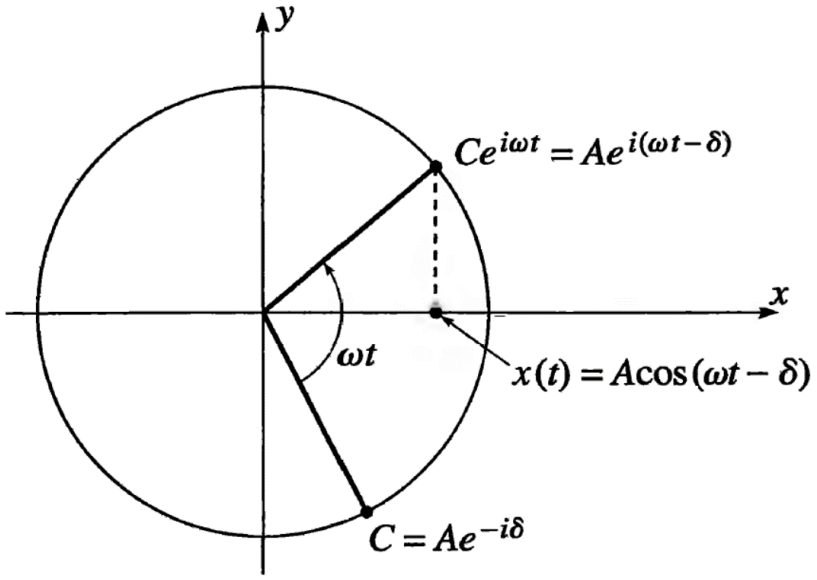
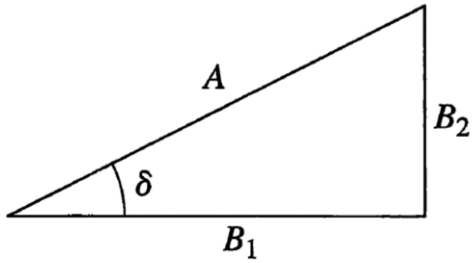


Figure 5.5 The position $x(t)$ of the oscillating cart is the real part of the complex number $Ae^{i(\omega t-\delta)}$. As the latter moves around the circle of radius A, the former oscillates back and forth on the x axis with amplitude A.

A phase-shifted cosine

$$x(t) = A\cos(\omega t - \delta)$$



SIMPLE HARMONIC MOTION: ENERGY

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

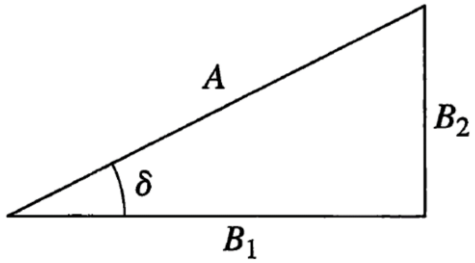
$$x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$$

Potential Energy

$$U = \tfrac{1}{2}kx^2 = \tfrac{1}{2}kA^2 \cos^2(\omega t - \delta)$$

A phase-shifted cosine

$$x(t) = A \cos(\omega t - \delta)$$



SIMPLE HARMONIC MOTION: ENERGY

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

$$x(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$$

Potential Energy

$$U = \tfrac{1}{2}kx^2 = \tfrac{1}{2}kA^2 \cos^2(\omega t - \delta)$$

Kinetic Energy

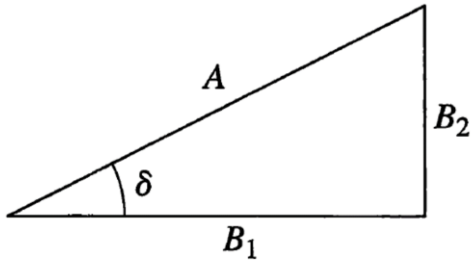
$$T = \tfrac{1}{2}m\dot{x}^2 = \tfrac{1}{2}m\omega^2 A^2 \sin^2(\omega t - \delta)$$

$$= \tfrac{1}{2}kA^2 \sin^2(\omega t - \delta)$$

$$E = T + U = \tfrac{1}{2}kA^2$$

A phase-shifted cosine

$$x(t) = A \cos(\omega t - \delta)$$



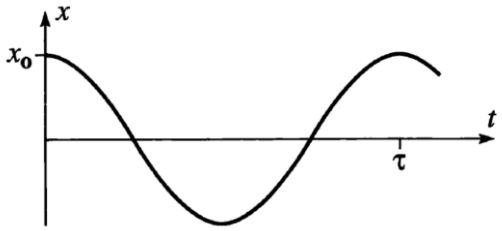
DAMPED OSCILLATIONS

Undamped

$$m\ddot{x} = F_x = -kx$$

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$x(t) = A\cos(\omega t - \delta)$$



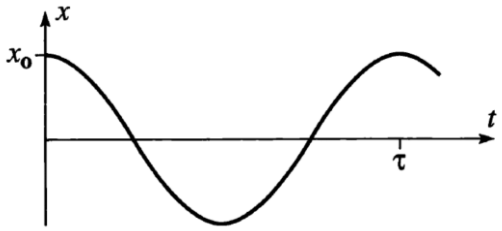
DAMPED OSCILLATIONS

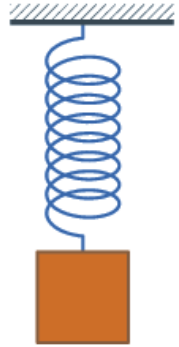
Undamped

$$m\ddot{x} = F_x = -kx$$

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$x(t) = A\cos(\omega t - \delta)$$





Damped

$$m\ddot{x} + b\dot{x} + kx = 0$$

Resistive force

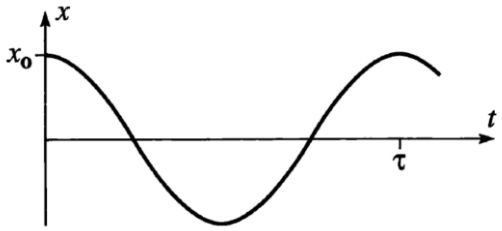
DAMPED OSCILLATIONS

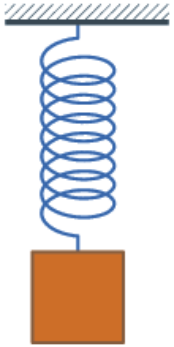
Undamped

$$m\ddot{x} = F_x = -kx$$

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$x(t) = A\cos(\omega t - \delta)$$





Damped

$$m\ddot{x} + b\dot{x} + kx = 0$$

Resistive force

A second-order, linear, homogeneous differential equation

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = 0$$

$$\frac{b}{m} = 2\beta$$

Damping constant

$$\omega_o = \sqrt{\frac{k}{m}}$$

Natural frequency

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

Zero

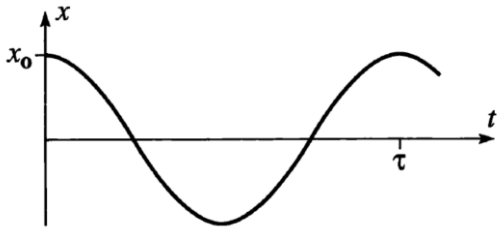
DAMPED OSCILLATIONS

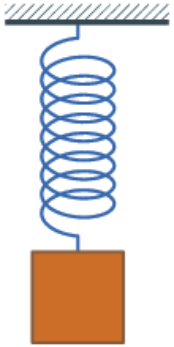
Undamped

$$m\ddot{x} = F_x = -kx$$

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$x(t) = A\cos(\omega t - \delta)$$





Damped

$$m\ddot{x} + b\dot{x} + kx = 0$$

Resistive force

A second-order, linear, homogeneous differential equation

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = 0$$

$$\frac{b}{m} = 2\beta$$

Damping constant

$$\omega_o = \sqrt{\frac{k}{m}}$$

Natural frequency

Let's guess a solution…

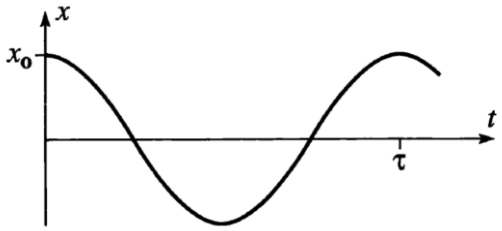
DAMPED OSCILLATIONS

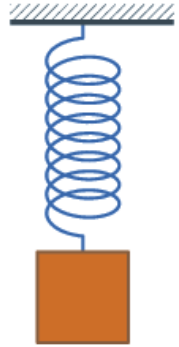
Undamped

$$m\ddot{x} = F_x = -kx$$

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$x(t) = A\cos(\omega t - \delta)$$





Damped

$$m\ddot{x} + b\dot{x} + kx = 0$$

Resistive force

A second-order, linear, homogeneous differential equation

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = 0$$

$$\frac{b}{m} = 2\beta$$

Damping constant

$$\omega_o = \sqrt{\frac{k}{m}}$$

Natural frequency

$$x(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

$$= e^{-\beta t}\left(C_1 e^{\sqrt{\beta^2 - \omega_o^2}\,t} + C_2 e^{-\sqrt{\beta^2 - \omega_o^2}\,t}\right)$$

DAMPED OSCILLATIONS

Undamped β is zero

$$x(t) = C_1 e^{i\omega_o t} + C_2 e^{-i\omega_o t}$$

Weak Damping $\beta < \omega_o$

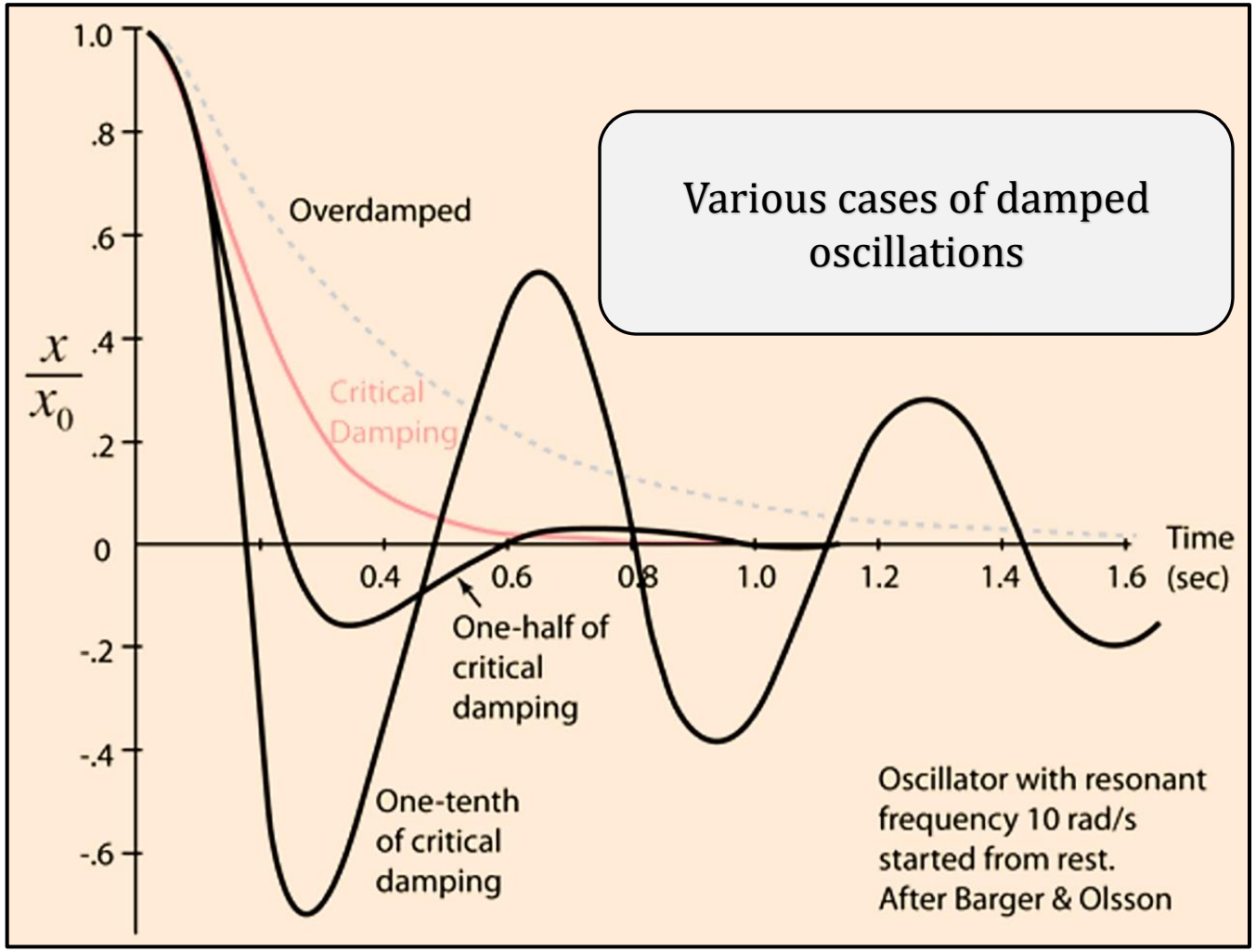
$$x(t) = Ae^{-\beta t}\cos(\omega_1 t - \delta)$$

Strong Damping $\beta > \omega_o$

$$x(t) = C_1 e^{-\left(\beta - \sqrt{\beta^2 - \omega_o^2}\right)t} + C_2 e^{-\left(\beta + \sqrt{\beta^2 - \omega_o^2}\right)t}$$

Critical Damping $\beta = \omega_o$

$$x(t) = C_1 e^{-\beta t} + C_2 t e^{-\beta t}$$



DAMPED OSCILLATIONS

Undamped β is zero

$$x(t) = C_1 e^{i\omega_o t} + C_2 e^{-i\omega_o t}$$

Weak Damping $\beta < \omega_o$

$$x(t) = Ae^{-\beta t}\cos(\omega_1 t - \delta)$$

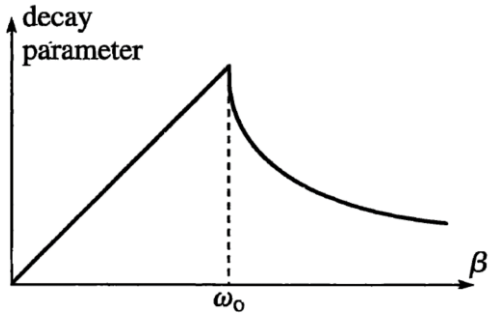
Strong Damping $\beta > \omega_o$

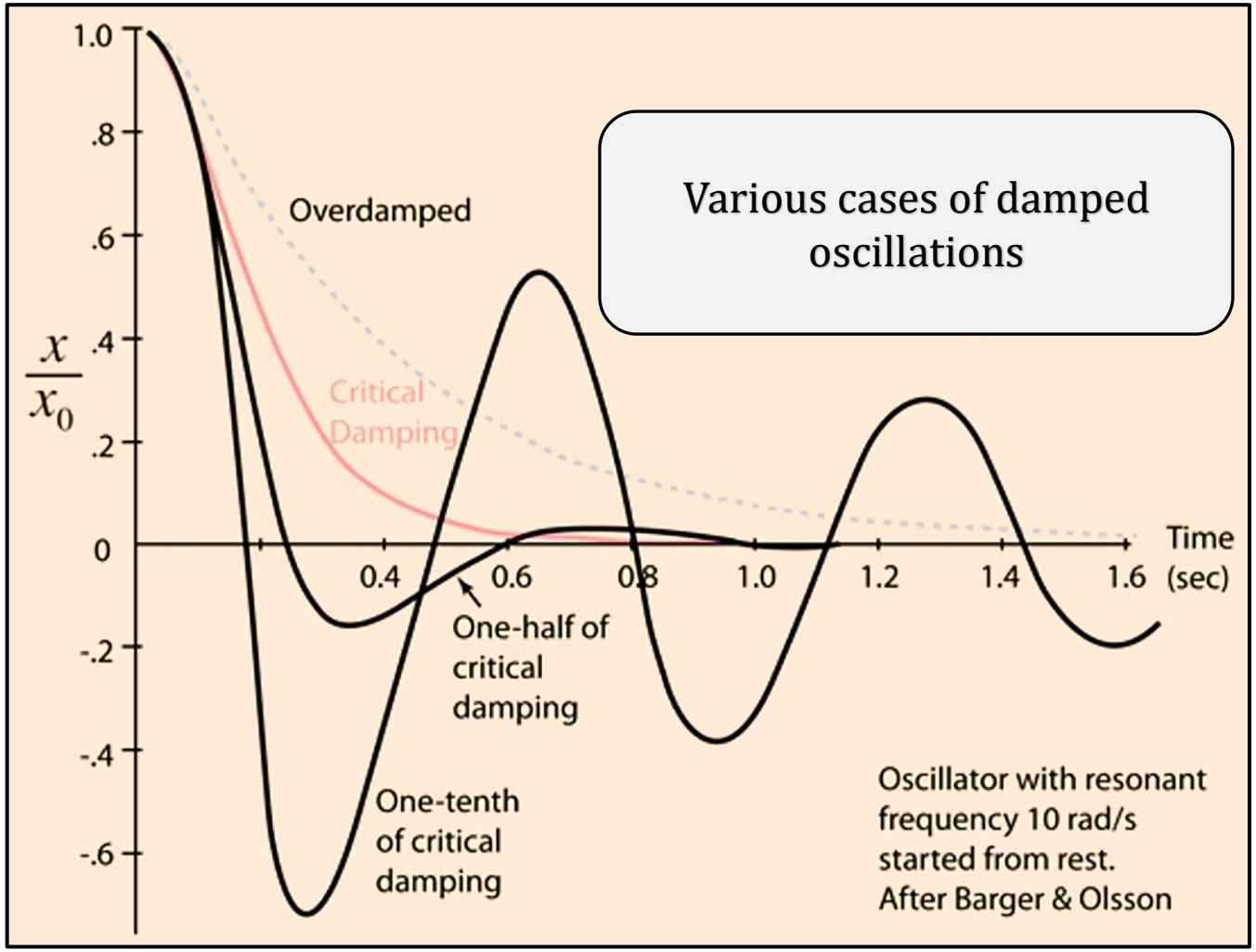
$$x(t) = C_1 e^{-\left(\beta - \sqrt{\beta^2 - \omega_o^2}\right)t} + C_2 e^{-\left(\beta + \sqrt{\beta^2 - \omega_o^2}\right)t}$$

Critical Damping $\beta = \omega_o$

$$x(t) = C_1 e^{-\beta t} + C_2 t e^{-\beta t}$$

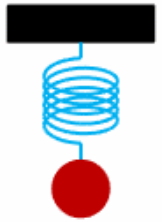
damping	β	decay parameter
none	$\beta = 0$	0
under	$\beta < \omega_o$	β
critical	$\beta = \omega_o$	β
over	$\beta > \omega_o$	$\beta - \sqrt{\beta^2 - \omega_o^2}$





DAMPED OSCILLATIONS: EXAMPLE 1

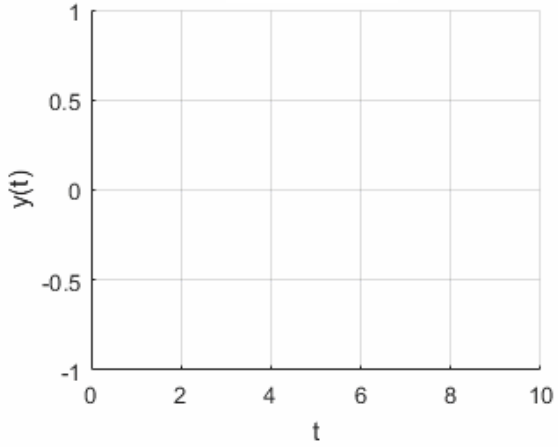
$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = 0$$





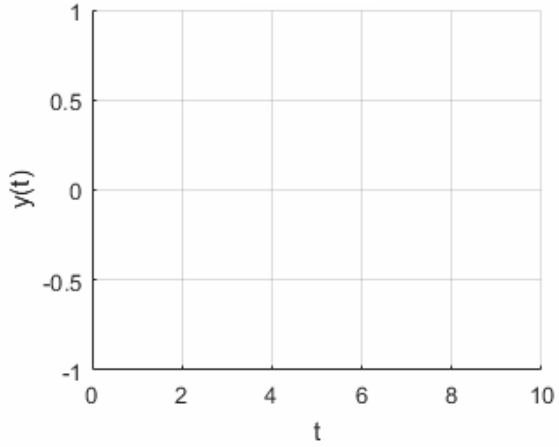


$\beta = 0.1$



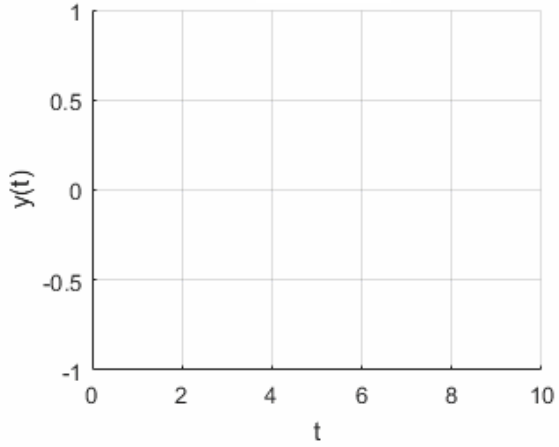
Underdamping

$\beta = 1$



Overdamping

$\beta = 3$



Overdamping

DAMPED OSCILLATIONS: EXAMPLE 2

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = 0$$

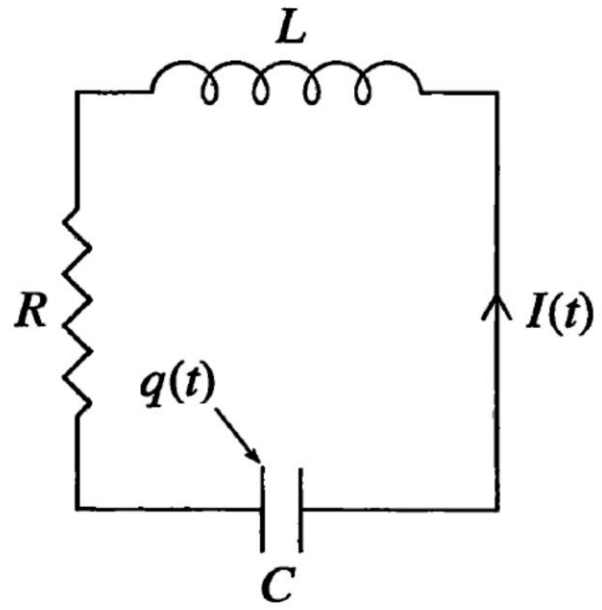
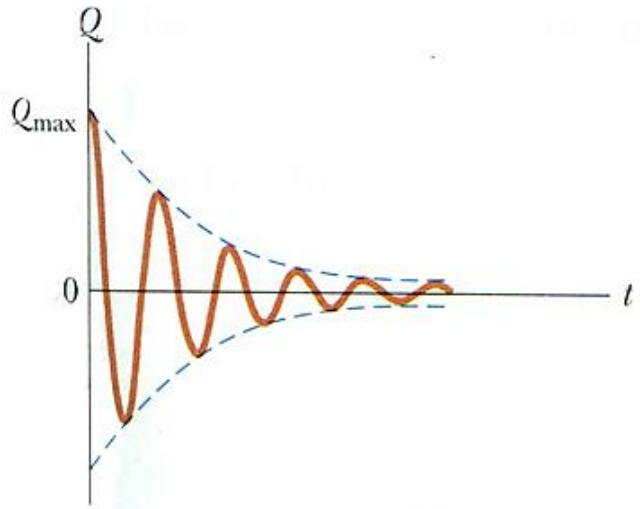


Figure 5.10 An LRC circuit.

DAMPED OSCILLATIONS: EXAMPLE 2

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = 0$$



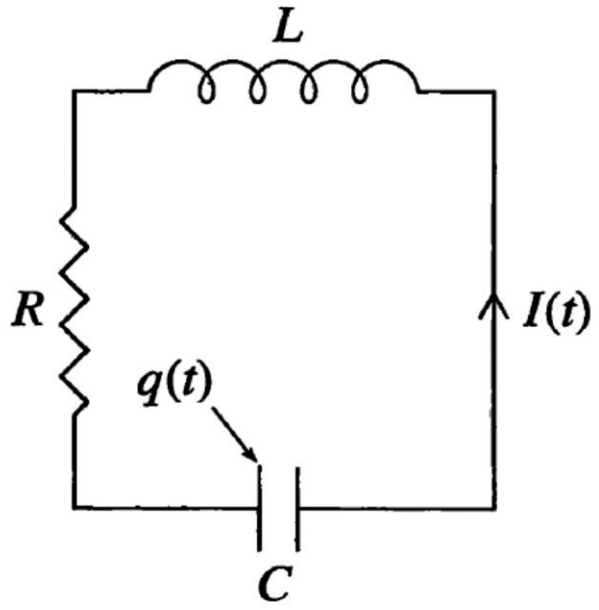


Figure 5.10 An LRC circuit.

$$I(t) = \dot{q}(t)$$

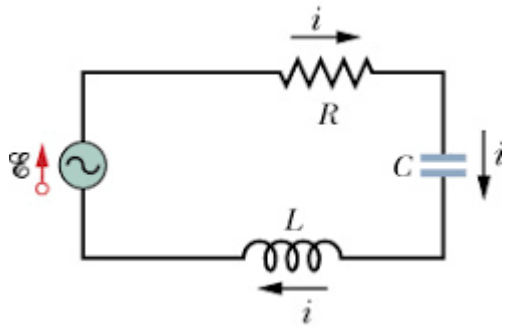
$$V_L = L\dot{I} = L\ddot{q}$$

$$V_R = RI = R\dot{q}$$

$$V_R = q/C$$

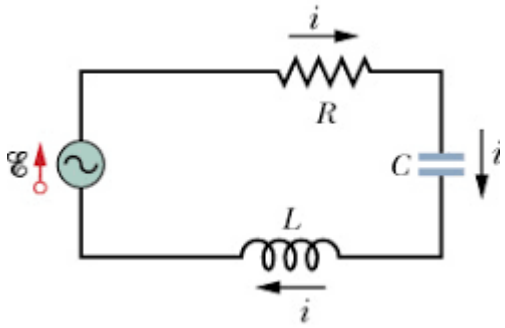
$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = 0$$

DRIVEN DAMPED OSCILLATIONS



$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = \mathcal{E}(t)$$

DRIVEN DAMPED OSCILLATIONS



$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = \mathcal{E}(t)$$

Inhomogeneous differential equation

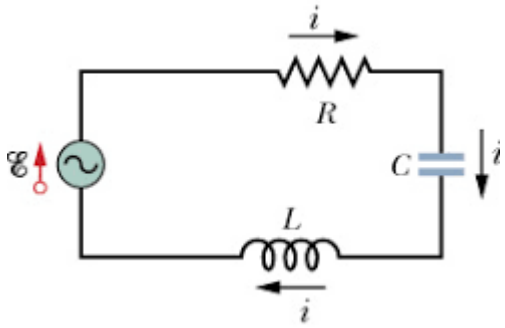
$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = f(t)$$

$$f(t) = \frac{F(t)}{m}$$

Particular solution

$x_p(t)$

DRIVEN DAMPED OSCILLATIONS



$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = \mathcal{E}(t)$$

Inhomogeneous differential equation

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = f(t)$$

$$f(t) = \frac{F(t)}{m}$$

Particular solution

$x_p(t)$

Homogeneous differential equation

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = 0$$

$$x(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

Homogenous solution

$x_h(t)$

General solution

$$x(t) = x_h(t) + x_p(t)$$

DRIVEN DAMPED OSCILLATIONS

Inhomogeneous differential equation

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = f(t) \qquad f(t) = \frac{F(t)}{m}$$

DRIVEN DAMPED OSCILLATIONS

Inhomogeneous differential equation

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = f(t) \qquad f(t) = \frac{F(t)}{m}$$

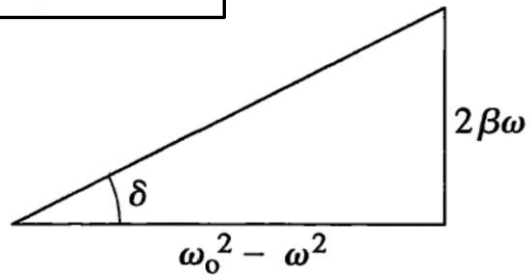
5.5 Driven Damped Oscillations

Complex Solutions for a Sinusoidal Driving Force

Particular solution for a sinusoidal driving force

$$f(t) = f_o \cos(\omega t) \qquad x_p(t) = A\cos(\omega t - \delta)$$

$$A^2 = \frac{f_o^2}{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2} \qquad \delta = \arctan\left(\frac{2\beta\omega}{\omega_o^2 - \omega^2}\right)$$



DRIVEN DAMPED OSCILLATIONS

Inhomogeneous differential equation

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = f(t) \qquad f(t) = \frac{F(t)}{m}$$

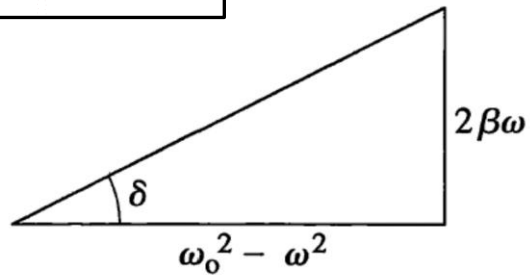
5.5 Driven Damped Oscillations

Complex Solutions for a Sinusoidal Driving Force

Particular solution for a sinusoidal driving force

$$f(t) = f_o \cos(\omega t) \qquad x_p(t) = A\cos(\omega t - \delta)$$

$$A^2 = \frac{f_o^2}{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2} \qquad \delta = \arctan\left(\frac{2\beta\omega}{\omega_o^2 - \omega^2}\right)$$



General solution

$$x(t) = A\cos(\omega t - \delta) + C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

$$x(t) = A\cos(\omega t - \delta) + A_{tr} e^{-\beta t}\cos(\omega_1 t - \delta_{tr})$$

Steady state solution:
Determined by the driving force

Transient solution:
Determined by initial conditions

DRIVEN DAMPED OSCILLATIONS

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = f(t).$$

$\omega_o = \sqrt{k/m}$ = natural frequency of undamped oscillator

$\omega_1 = \sqrt{\omega_o^2 - \beta^2}$ = frequency of damped oscillator

DRIVEN DAMPED OSCILLATIONS

$$A^2 = \frac{f_o^2}{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = f(t).$$

$\omega_o = \sqrt{k/m}$ = natural frequency of undamped oscillator

$\omega_1 = \sqrt{\omega_o^2 - \beta^2}$ = frequency of damped oscillator

ω = frequency of driving force

$\omega_2 = \sqrt{\omega_o^2 - 2\beta^2}$ = value of ω at which response is maximum.

Resonance

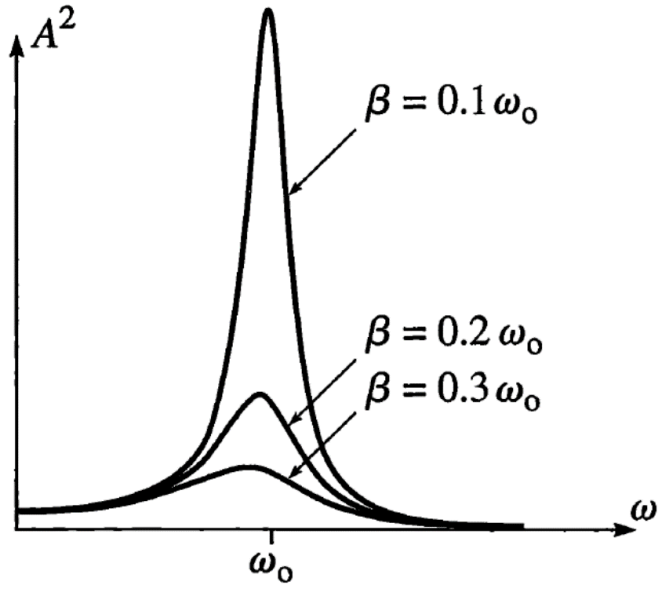


Figure 5.17 The amplitude for driven oscillations as a function of the driving frequency ω for three different values of the damping constant β. Note how as β decreases the resonance peak gets higher and sharper.

DRIVEN DAMPED OSCILLATIONS

$$A^2 = \frac{f_o^2}{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

$$\ddot{x} + 2\beta\dot{x} + \omega_o^2 x = f(t).$$

$\omega_o = \sqrt{k/m}$ = natural frequency of undamped oscillator

$\omega_1 = \sqrt{\omega_o^2 - \beta^2}$ = frequency of damped oscillator

ω = frequency of driving force

$\omega_2 = \sqrt{\omega_o^2 - 2\beta^2}$ = value of ω at which response is maximum.

when $\beta \ll \omega_o$ ⇨ $\omega_o \approx \omega$

$$A_{max} \approx \frac{f_o}{2\beta\omega_o}$$

Resonance

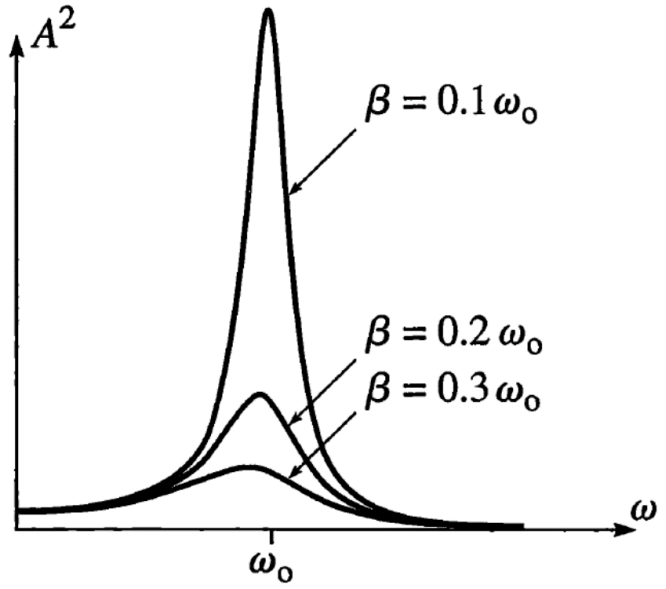


Figure 5.17 The amplitude for driven oscillations as a function of the driving frequency ω for three different values of the damping constant β. Note how as β decreases the resonance peak gets higher and sharper.

DRIVEN DAMPED OSCILLATIONS

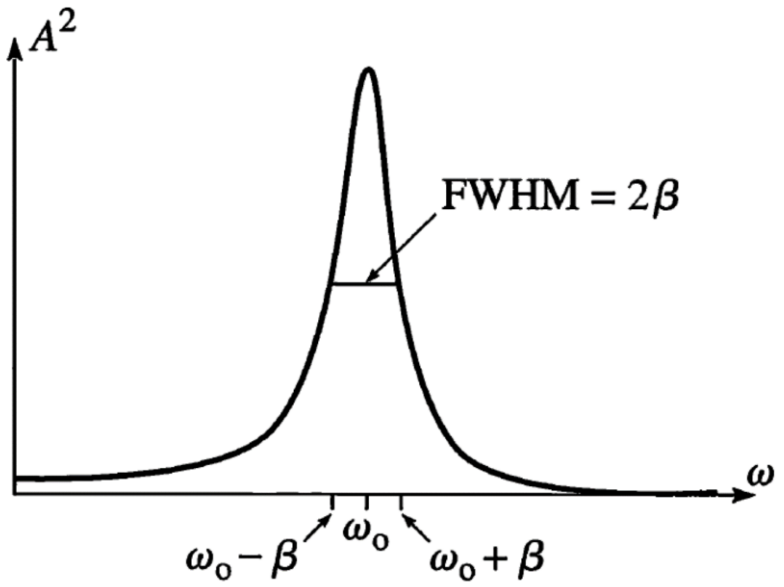
$$A^2 = \frac{f_o^2}{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

when $\beta \ll \omega_o$ ⇨ $\omega_o \approx \omega$

$$A_{max} \approx \frac{f_o}{2\beta\omega_o}$$

ω = frequency of driving force

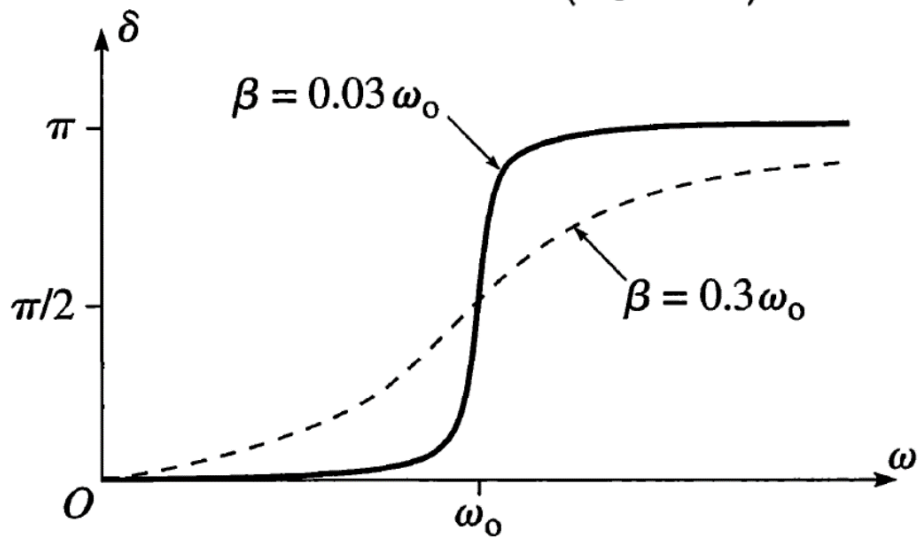
$\omega_2 = \sqrt{\omega_o^2 - 2\beta^2}$ = value of ω at which response is maximum.



Resonance

Quality factor $Q = \frac{\omega_o}{2\beta} = \pi \frac{1/\beta}{2\pi/\omega_o} = \pi \frac{\text{decay time}}{\text{period}}$

Phase $\delta = \arctan\left(\frac{2\beta\omega}{\omega_o^2 - \omega^2}\right)$



where $\omega = \omega_o$, the argument of the arctangent in is infinite, so $\delta = \pi/2$